

PROJECT BUTTON HR–V0 ELECTRICAL V3 OBSERVATION–INTEGRATED CANDIDATE

P1.15 limiter chain plus R211/P0.5 open–drain observation carrier and R204 Pi interface; zero safety credit.

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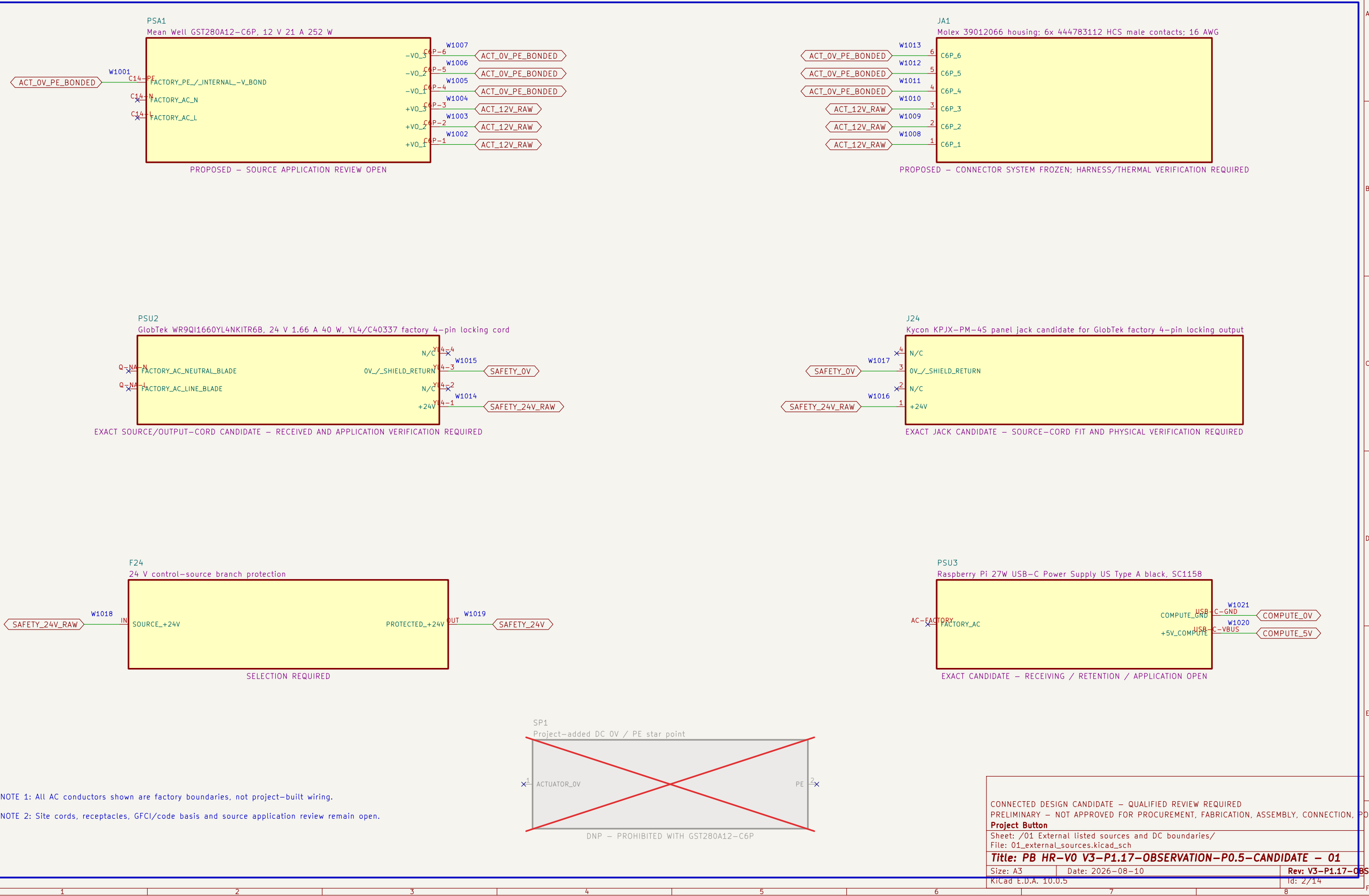
File: 12_watchdog_pcb_test_access.kicad_sch

13 Runtime diagnostic observation interfaces

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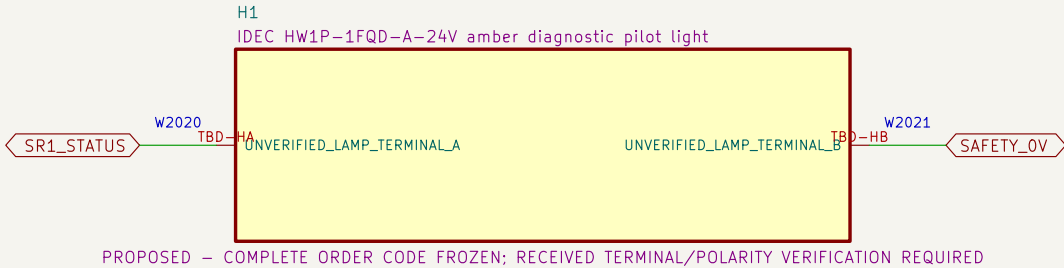
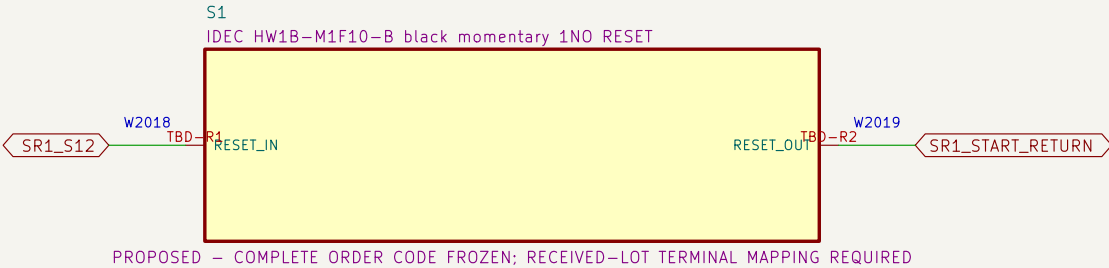
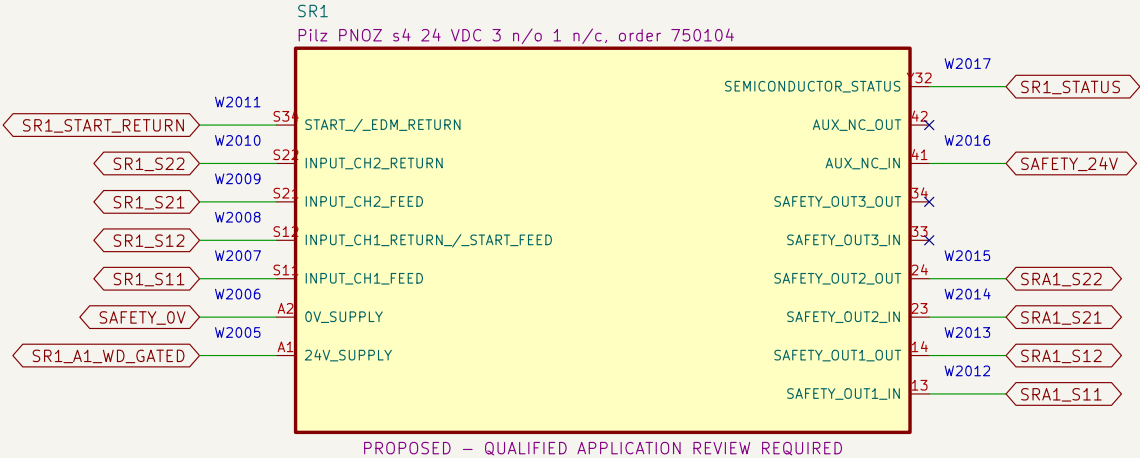
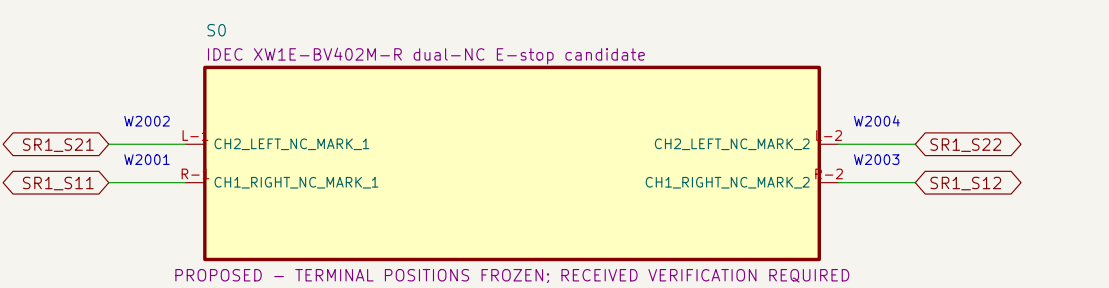
01 External listed sources and DC boundaries

Factory-sealed adapters eliminate project-built mains wiring.



02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains only one S0 NC contact; RESET cannot energize K1/K2.

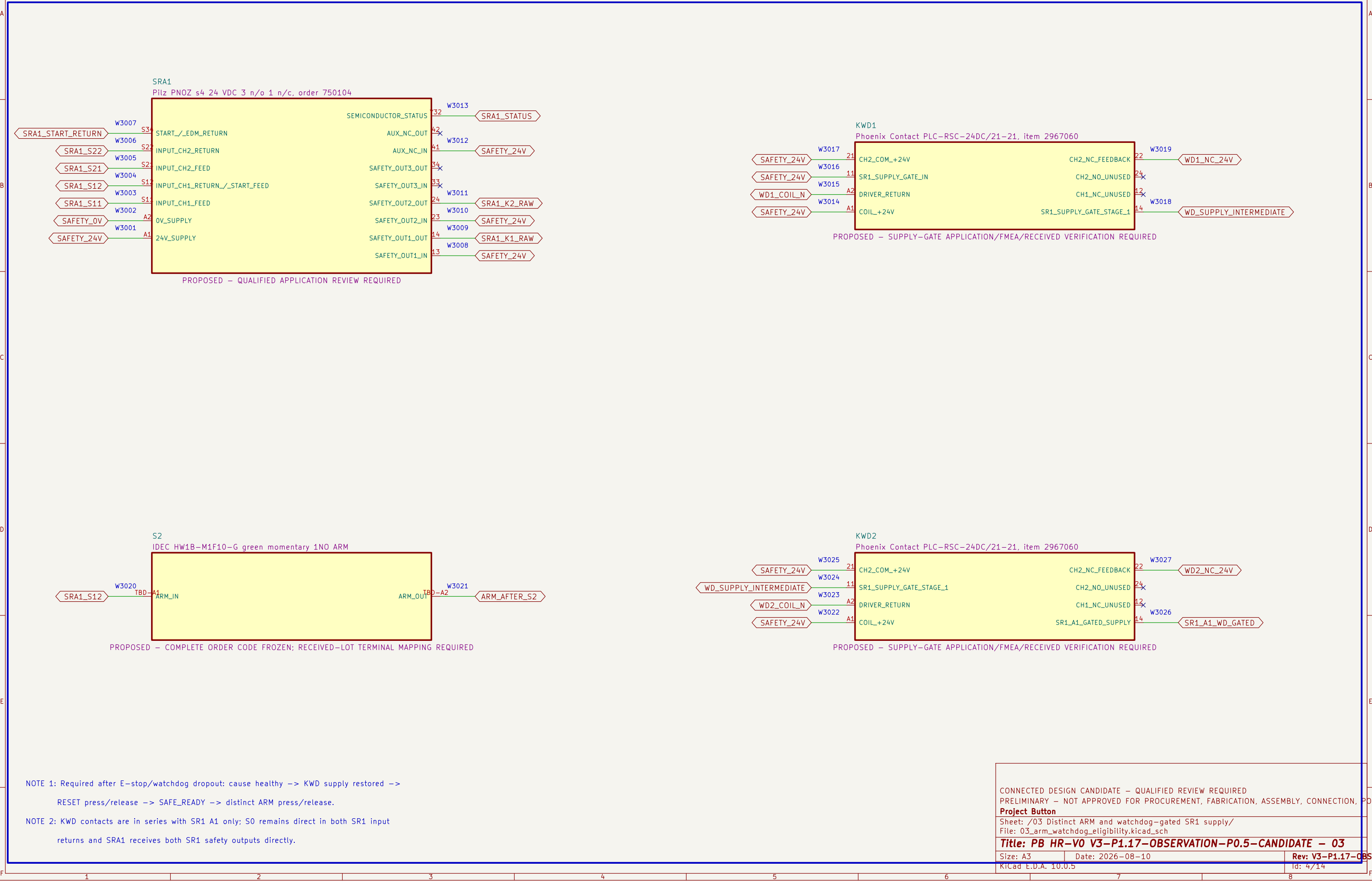


NOTE 1: S0 directly completes both SR1 input returns; ordinary KWD terminals are absent from both loops.

NOTE 2: Heartbeat loss removes SR1 A1 through the separate two-contact supply gate; recovery alone cannot restore the monitored RESET stage.

03 Distinct ARM and watchdog-gated SR1 supply

Two ordinary KWD contacts gate SR1 A1; SRA1 still requires SR1 outputs, EDM proof and a new ARM action.



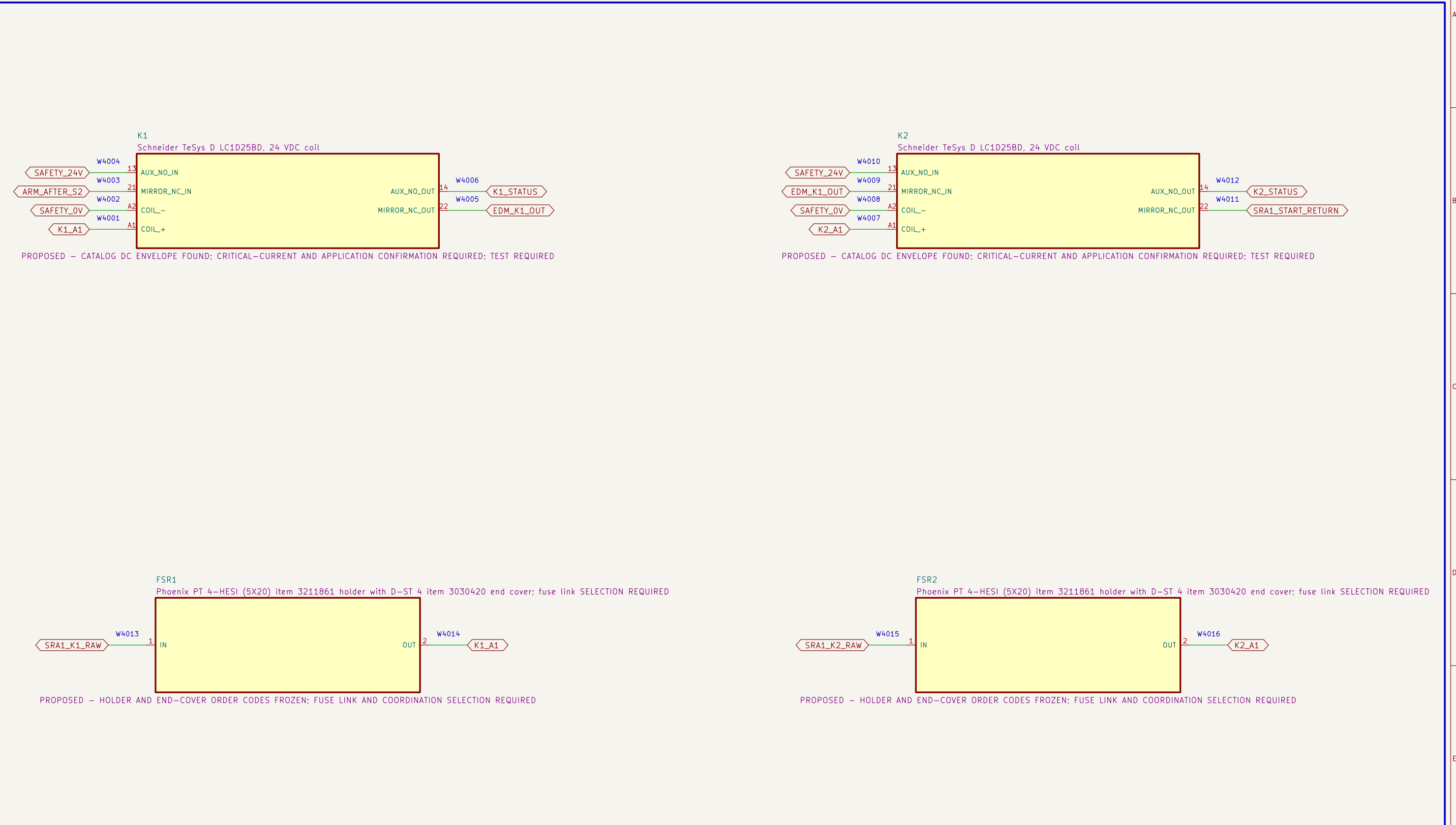
NOTE 1: Required after E-stop/watchdog dropout: cause healthy -> KWD supply restored ->
RESET press/release -> SAFE_READY -> distinct ARM press/release.

NOTE 2: KWD contacts are in series with SR1 A1 only; S0 remains direct in both SR1 input
returns and SRA1 receives both SR1 safety outputs directly.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION
Project Button
Sheet: /03 Distinct ARM and watchdog-gated SR1 supply/
File: 03_arm_watchdog_eligibility.kicad_sch
Title: PB HR-V0 V3-P1.17-OBSERVATION-P0.5-CANDIDATE – 03
Size: A3
Date: 2026-08-10
Kicad E.D.A. 10.0.5
Rev: V3-P1.17-OBSERVATION
Id: 4/14

04 Contactor coils, mirror contacts and EDM

K1 and K2 are distinct final elements; their mirror contacts form the monitored restart return.



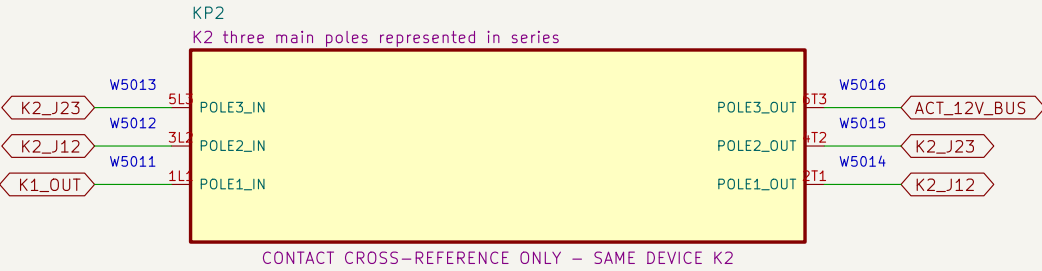
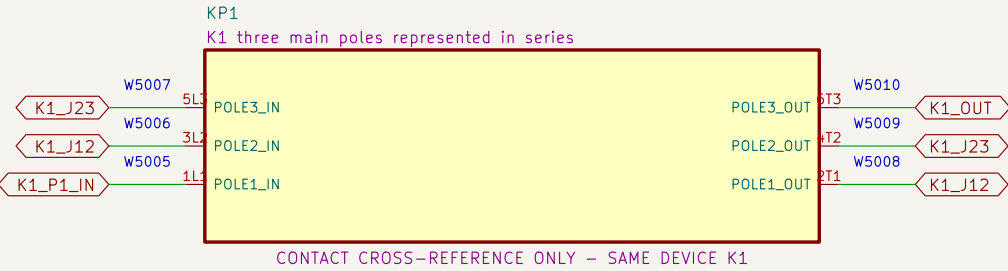
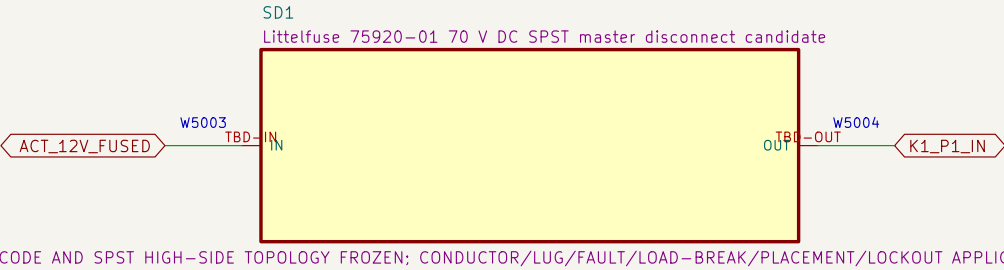
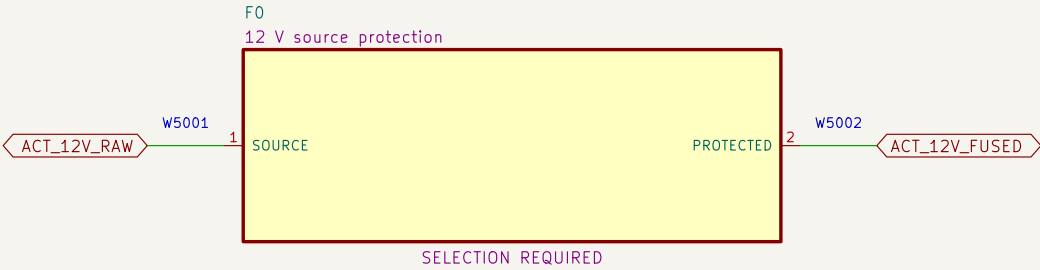
NOTE 1: SRA1 outputs are separately protected before K1 and K2 coils. FSR1/FSR2 use proposed Phoenix PT 4-HESI item 3211861 holders with D-ST 4 item 3030420 end cover; both fuse links, grouping and coordination remain SELECTION REQUIRED.

NOTE 2: Loaded DC interruption, suppression behavior, mirror-contact use and coordination require qualified review.

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PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION		
Project Button		
Sheet: /04 Contactor coils, mirror contacts and EDM/		
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Size: A3	Date: 2026-08-10	Rev: V3-P1.17-OBSERVATION
KiCad E.D.A. 10.0.5	Id: 5/14	

05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



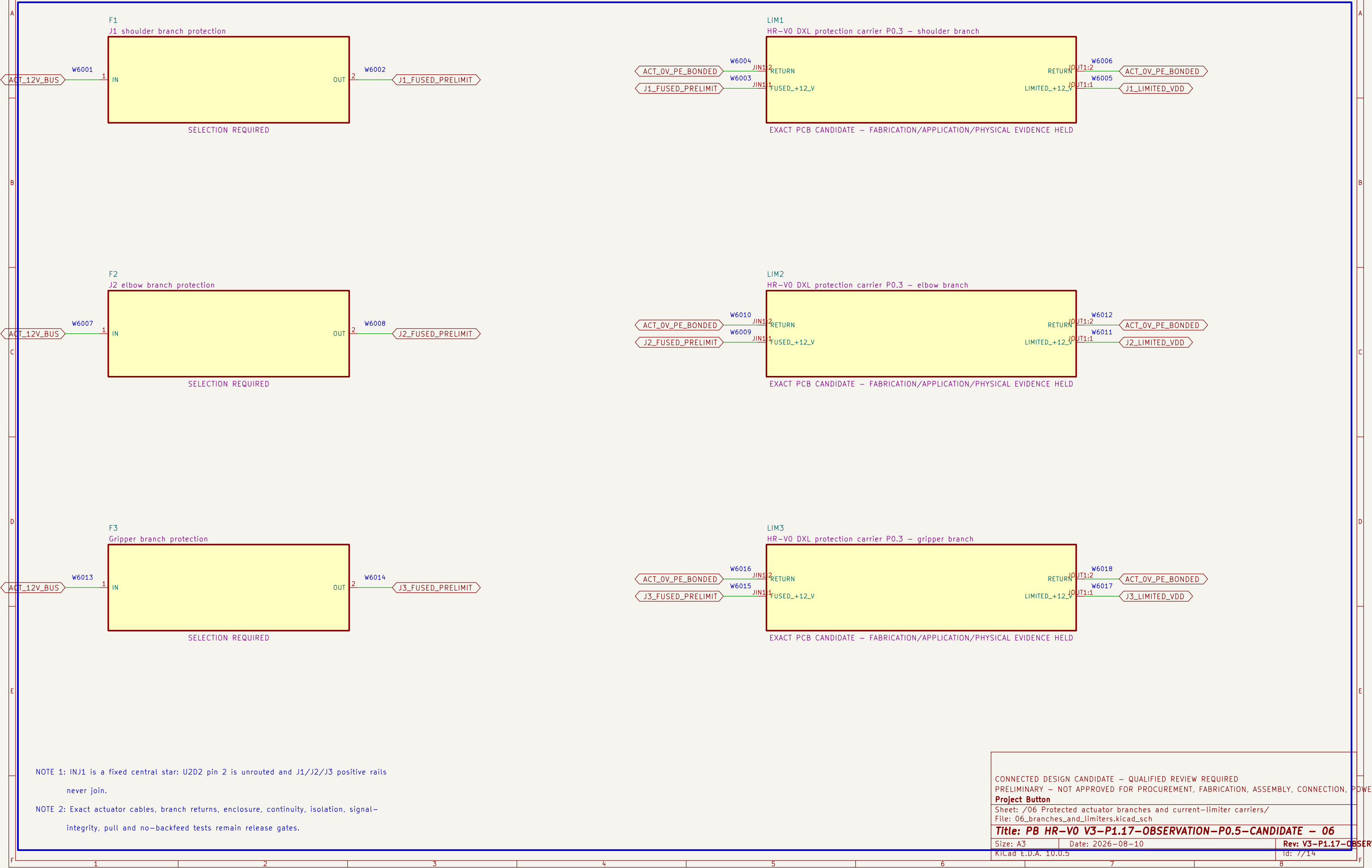
NOTE 1: Series jumpers: K1 2T1–>3L2, 4T2–>5L3, 6T3–>K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

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PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION		
Project Button		
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KiCad E.D.A. 10.0.5		Id: 6/14

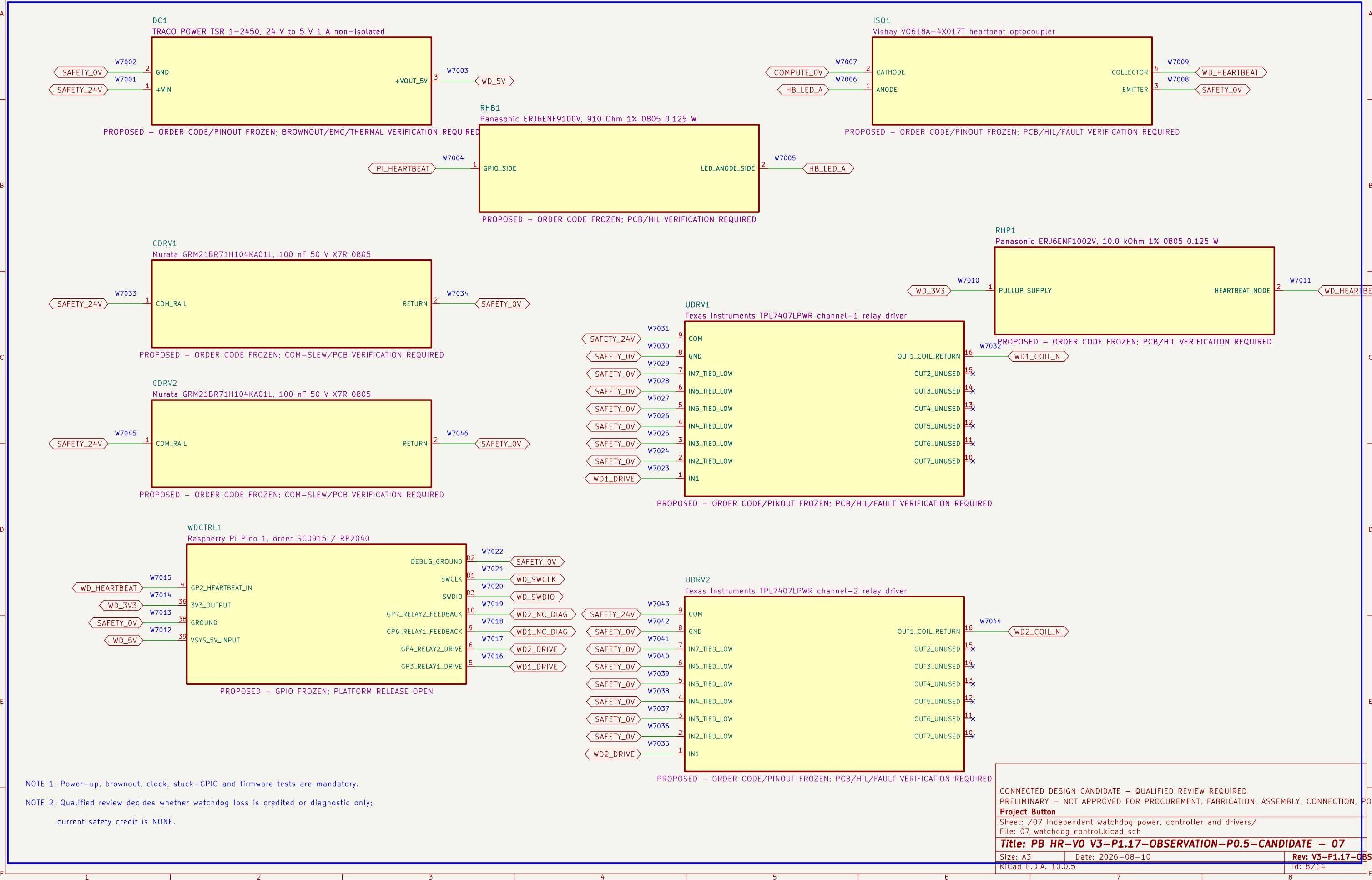
06 Protected actuator branches and current-limiter carriers

Each fuse output and carrier output has a distinct positive rail; physical protection performance remains unproved.



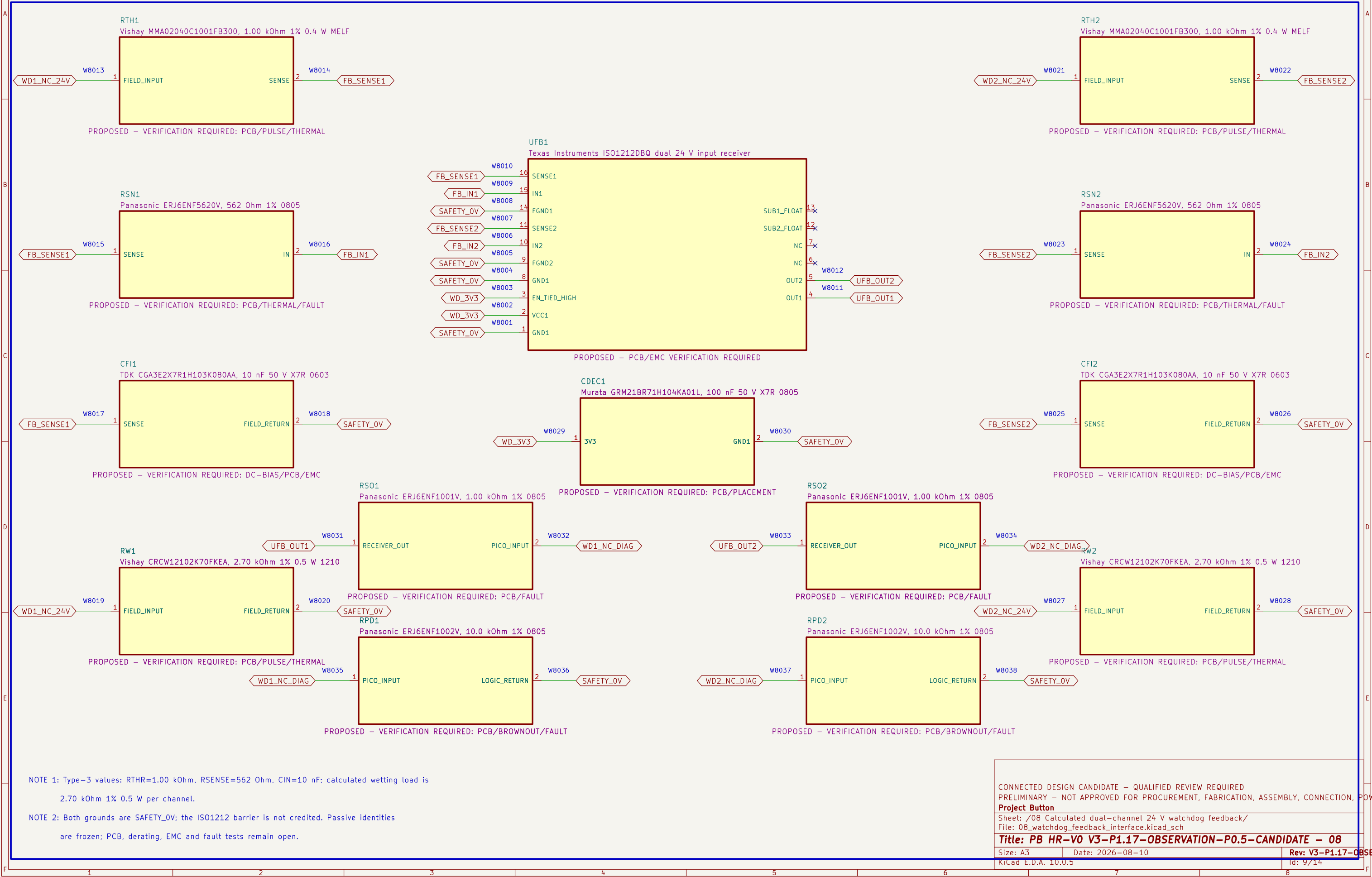
07 Independent watchdog power, controller and drivers

The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.



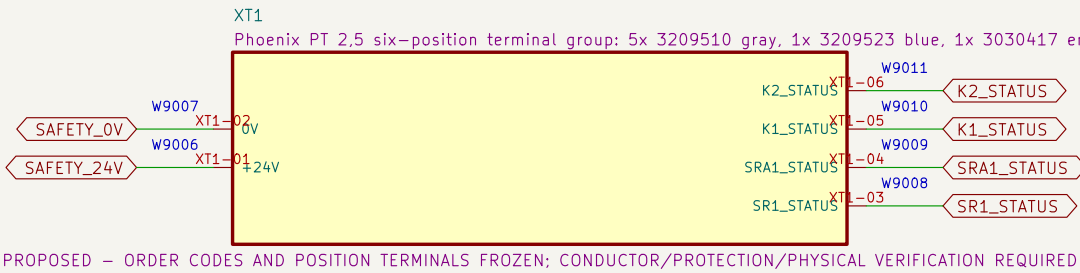
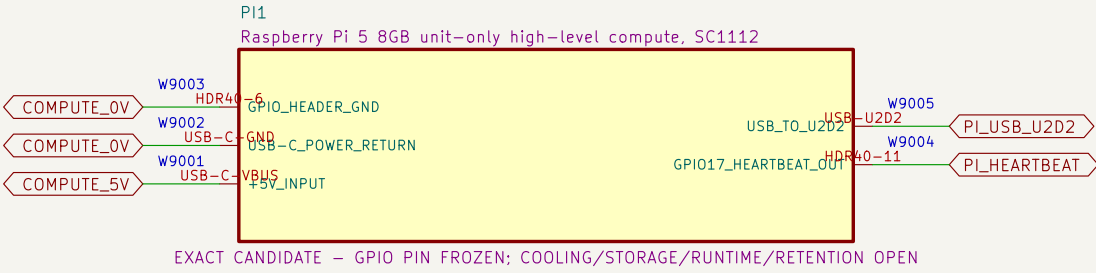
08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



09 Compute and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



NOTE 1: No installed debug connector exists; watchdog programming uses only TP15/TP16/TP2 under a future controlled unpowered-fixture procedure.

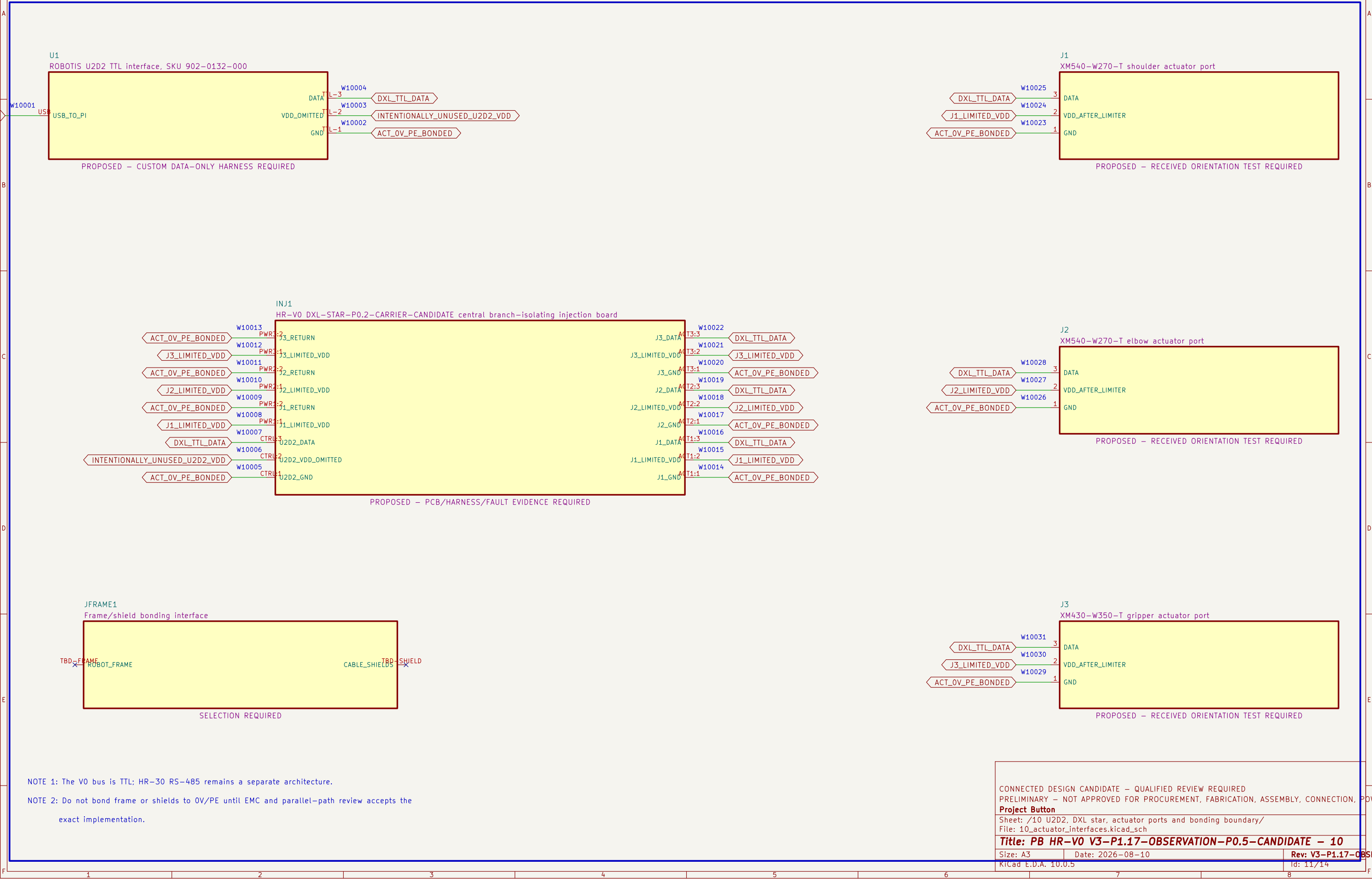
NOTE 2: Debug connection, halt, flash and disconnect must not enable outputs, back-power the board or bypass a protective function.

NOTE 3: Heartbeat cable, GPIO runtime, timing, terminal family, markers, conductor range, torque and enclosure layout remain selection required.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION		
Project Button		
Sheet: /09 Compute and control terminals/ File: 09_compute_and_control_terminals.kicad_sch		
Title: PB HR-V0 V3-P1.17-OBSERVATION-P0.5-CANDIDATE – 09		
Size: A3	Date: 2026-08-10	Rev: V3-P1.17-OBSERVATION
KiCad E.D.A. 10.0.5	Id: 10/14	

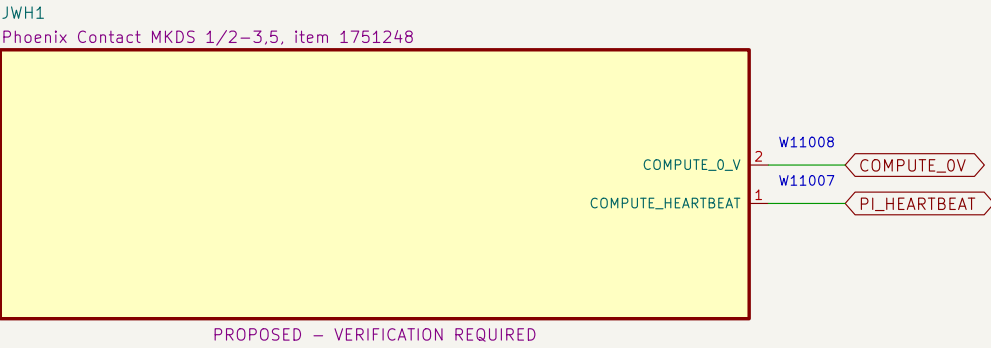
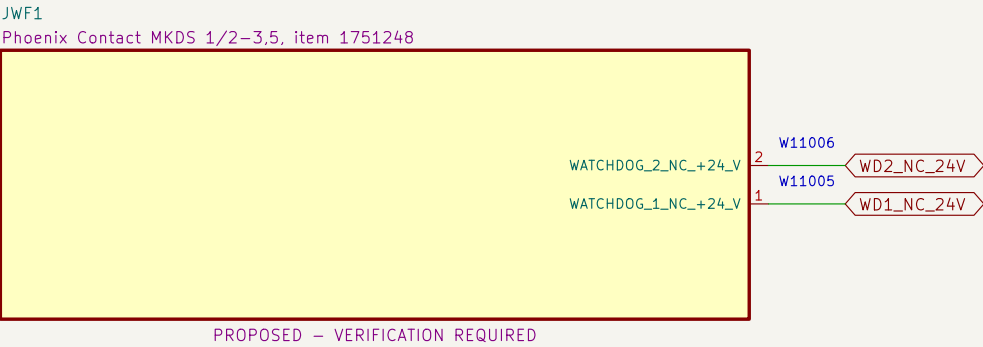
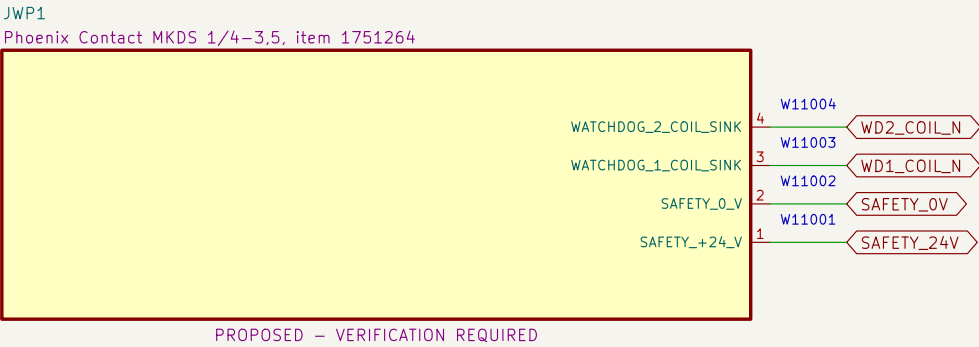
10 U2D2, DXL star, actuator ports and bonding boundary

U2D2 pin 2 is omitted; DXL–STAR accepts three distinct limited positive rails and routes DATA/return.



11 Watchdog PCB external connectors

Exact PCB terminal–block candidates and project pin allocations define the board boundary; harness release remains open.



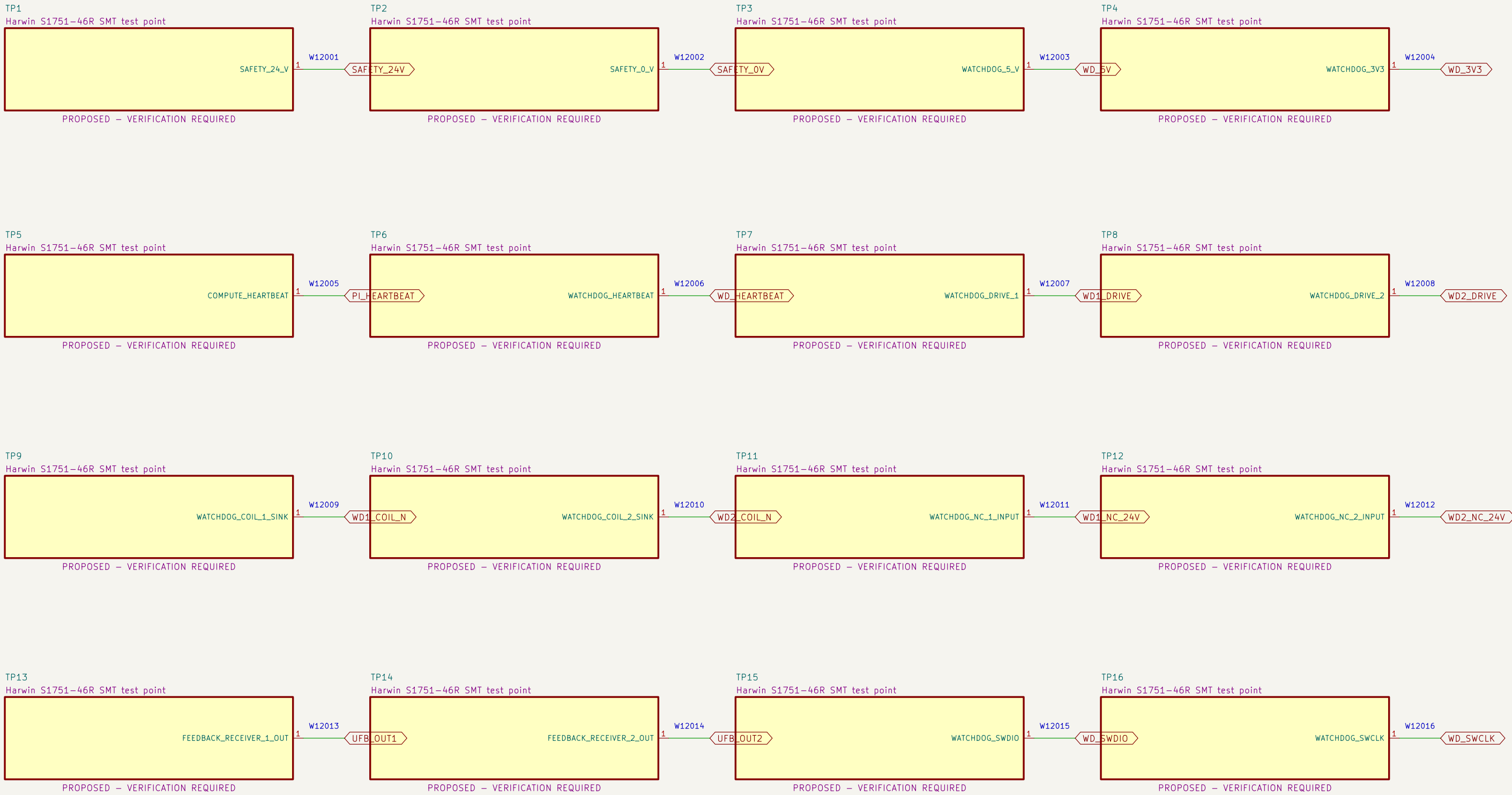
NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

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PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION		
Project Button		
Sheet: /11 Watchdog PCB external connectors/ File: 11_watchdog_pcb_connectors.kicad_sch		
Title: PB HR-V0 V3-P1.17-OBSERVATION-P0.5-CANDIDATE – 11		
Size: A3	Date: 2026-08-10	Rev: V3-P1.17-OBSERVATION
KiCad E.D.A. 10.0.5		Id: 12/14

12 Watchdog PCB test access

Exact clip-compatible test terminals expose controlled nodes without granting safety or operating authority.

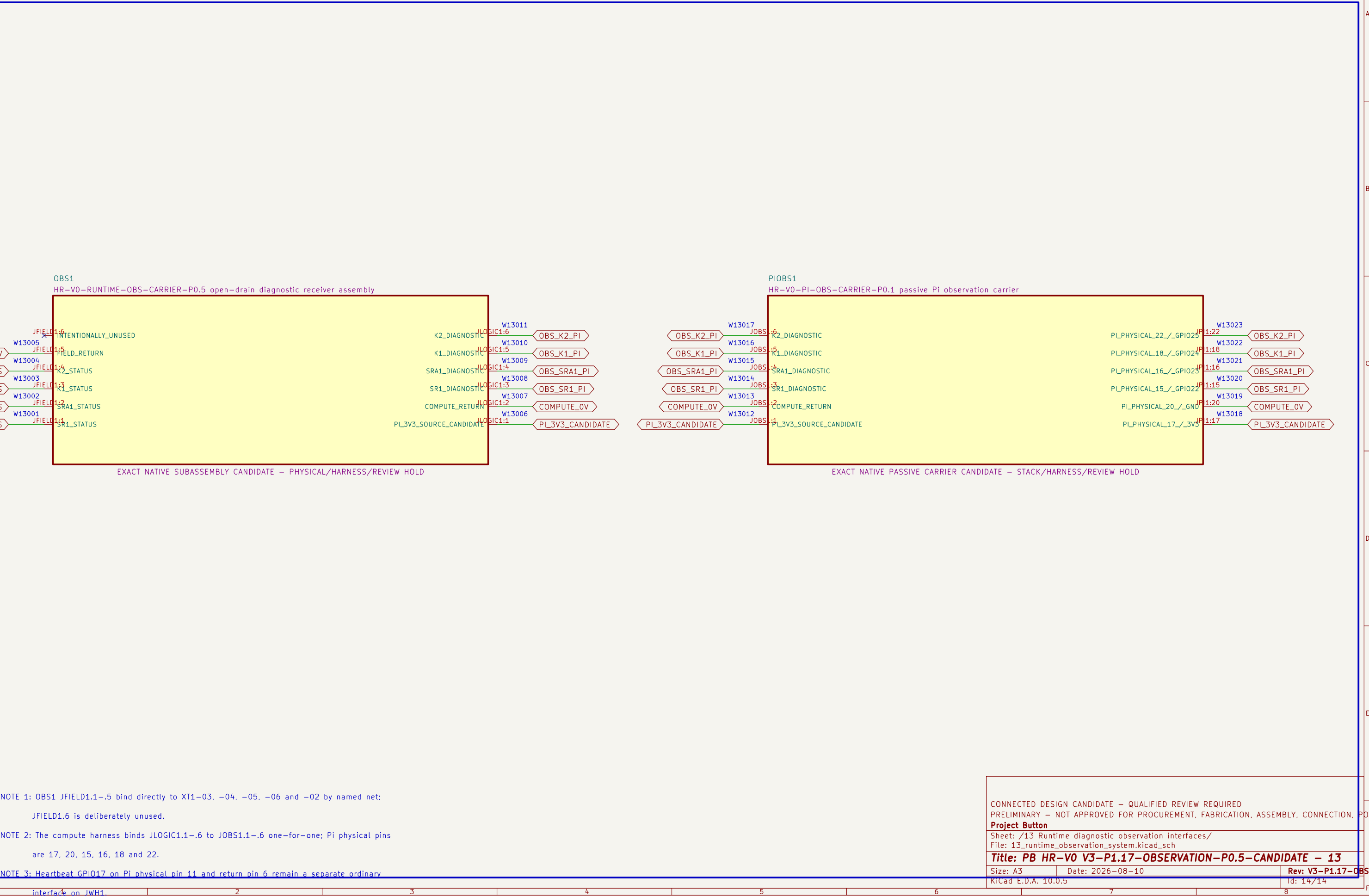


NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.

13 Runtime diagnostic observation interfaces

XT1 status conductors feed the R211 isolated receiver; R204 carries six compute–side nets to exact Pi physical pins.



NOTE 1: OBS1 JFIELD1.1–.5 bind directly to XT1–03, –04, –05, –06 and –02 by named net;

JFIELD1.6 is deliberately unused.

NOTE 2: The compute harness binds JLOGIC1.1–.6 to JOBS1.1–.6 one-for-one; Pi physical pins are 17, 20, 15, 16, 18 and 22.

NOTE 3: Heartbeat GPIO17 on Pi physical pin 11 and return pin 6 remain a separate ordinary

NOTE 4: The observation chain is diagnostic only: it cannot command, restore, latch or

energize, and is not a functional safety critical